MSG No. 23.50 $I2221'$ [Type II, orthorhombic]

Table 1: Wyckoff site: `2a`, site symmetry: `2221'`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,9,10,11,12]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5,6,7,8,13,14,15,16]`

Table 2: Wyckoff site: `2b`, site symmetry: `2221'`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	`[1,2,3,4,9,10,11,12]`
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[5,6,7,8,13,14,15,16]`

Table 3: Wyckoff site: `2c`, site symmetry: `2221'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,3,4,9,10,11,12]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[5,6,7,8,13,14,15,16]`

Table 4: Wyckoff site: `2d`, site symmetry: `2221'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,2,3,4,9,10,11,12]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,6,7,8,13,14,15,16]`

Table 5: Wyckoff site: `4e`, site symmetry: `2..1'`

No.	position	mapping
1	$[x, 0, 0]$	`[1,2,9,10]`
2	$[-x, 0, 0]$	`[3,4,11,12]`
3	$\left[x + \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5,6,13,14]`
4	$\left[\frac{1}{2} - x, \frac{1}{2}, \frac{1}{2}\right]$	`[7,8,15,16]`

Table 6: Wyckoff site: `4f`, site symmetry: `2..1'`

No.	position	mapping
1	$\left[x, 0, \frac{1}{2}\right]$	`[1,2,9,10]`
2	$\left[-x, 0, \frac{1}{2}\right]$	`[3,4,11,12]`
3	$\left[x + \frac{1}{2}, \frac{1}{2}, 0\right]$	`[5,6,13,14]`
4	$\left[\frac{1}{2} - x, \frac{1}{2}, 0\right]$	`[7,8,15,16]`

Table 7: Wyckoff site: `4g`, site symmetry: `.2.1'`

No.	position	mapping
1	$[0,\ y,\ 0]$	`[1,3,9,11]`
2	$[0,\ -y,\ 0]$	`[2,4,10,12]`
3	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[5,7,13,15]`
4	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[6,8,14,16]`

Table 8: Wyckoff site: `4h`, site symmetry: `.2.1'`

No.	position	mapping
1	$\left[\frac{1}{2},\ y,\ 0\right]$	`[1,3,9,11]`
2	$\left[\frac{1}{2},\ -y,\ 0\right]$	`[2,4,10,12]`
3	$\left[0,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[5,7,13,15]`
4	$\left[0,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[6,8,14,16]`

Table 9: Wyckoff site: `4i`, site symmetry: `..21'`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,4,9,12]`
2	$[0,\ 0,\ -z]$	`[2,3,10,11]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[5,8,13,16]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[6,7,14,15]`

Table 10: Wyckoff site: `4j`, site symmetry: `..21'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,4,9,12]`
2	$\left[0,\ \frac{1}{2},\ -z\right]$	`[2,3,10,11]`
3	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[5,8,13,16]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[6,7,14,15]`

Table 11: Wyckoff site: `8k`, site symmetry: `11'`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1,9]`
2	$[x,\ -y,\ -z]$	`[2,10]`
3	$[-x,\ y,\ -z]$	`[3,11]`
4	$[-x,\ -y,\ z]$	`[4,12]`
5	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[5,13]`
6	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	`[6,14]`
7	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[7,15]`
8	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[8,16]`